Athena Elvey

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Education

Rensselaer Polytechnic Institute Troy, NY

B.S. Computer Science & Cognitive Science *Expected Graduation: Dec. 2028*

- **Honors:** AP Scholar with Distinction, Garnet D. Baltimore Scholarship, National African-American Recognition Program
- **Relevant Coursework:** Foundations of Computer Science, Data Structures, Linear Algebra, Cognitive Neuroscience, Computer Architecture and Operating Systems

Technical Skills

Languages: Java, Python, C++, C, JavaScript, HTML/CSS, Rust
Frameworks & Tools: React, Vite, React Router, Tailwind CSS, FastAPI, SQLAlchemy, Alembic, Pydantic, PostgreSQL, Docker, Git
Data & ML: Pandas, NumPy, scikit-learn, TensorFlow/Keras, Altair, Matplotlib

Projects

Chronova | *React, Vite, FastAPI, PostgreSQL, SQLAlchemy, Three.js, Plotly* github.com/athenaelvey/chronova

- Architected a full-stack application over the ATNF Pulsar Catalogue: a React/Vite frontend, a FastAPI backend with Pydantic-validated request/response models, and a containerized (Docker) PostgreSQL database with Alembic-managed schema migrations
- Designed a dynamic query builder translating arbitrary user filter conditions into SQLAlchemy expressions via a REST API for flexible, structured search, and automated live data ingestion via psrqpy
- Built interactive 3D and data visualizations with Three.js and Plotly, turning raw catalogue data into an explorable interface

Daily Cognitive Load Tracker | *Python, Streamlit, Pandas, Altair, NumPy* daily-cognitive-load-tracker.streamlit.app

- Built and deployed a wellness tracking app for logging and visualizing daily stress, energy, and focus levels with streak tracking and pattern-based alerts
- Implemented NumPy linear regression for real-time trend lines across metrics, with CSV persistence and a live Streamlit Cloud deployment
- Designed interactive Altair visualizations to surface focus, stress, and energy trends over time

Hospital Length-of-Stay Prediction Model | *Python, Pandas, scikit-learn, TensorFlow/Keras*

- Analyzed healthcare infrastructure data on a team to identify relationships between medical equipment availability and patient length of stay; benchmarked linear regression, polynomial regression, and a tuned neural network (TensorFlow/Keras) to optimize predictive performance

Experience

Creative Mortgage Solutions Sep. 2026 - Present

Operations Support Intern

- Maintain and update lender and client databases, ensuring information is accurate, properly categorized, and accessible while monitoring records and outstanding requirements to keep files moving toward closing
- Utilize digital platforms and internal systems to manage client files, coordinate workflows, and communicate task status with the operations team through Microsoft Teams
- Support day-to-day mortgage operations by organizing documentation, tracking open items, and flagging gaps so the team can respond to lenders and clients quickly

America Reads, America Counts, Tutor Time & STEP Sep. 2025 - Present

Tutor *Troy, NY*

- Tutored 20+ grade school and high school students in literacy, math, and computer science, improving test scores

Leadership and Organizations

National Society of Black Engineers – RPI Chapter Sep. 2026 - Present

Communications Committee *Troy, NY*

- Support NSBE's communications efforts by coordinating and sharing information about events, programs, and opportunities with members and the broader RPI community

SEO Career Jul. 2026 - Present

SEO EDGE Participant *New York, NY*

- Built a foundation in core business concepts including strategy, finance, and professional communication through targeted coaching and structured training

RPI Undergraduate Council / Class of 2029 Council Oct. 2025 - Mar. 2026

Class of 2029 Vice President *Troy, NY*

- Represented 1,600 students' interests by contributing to freshman well-being initiatives on campus